LESSON 9 • TERM 4 • LEVEL 1

# The Problem-Solving Toolbox

Pick the tool, then check it

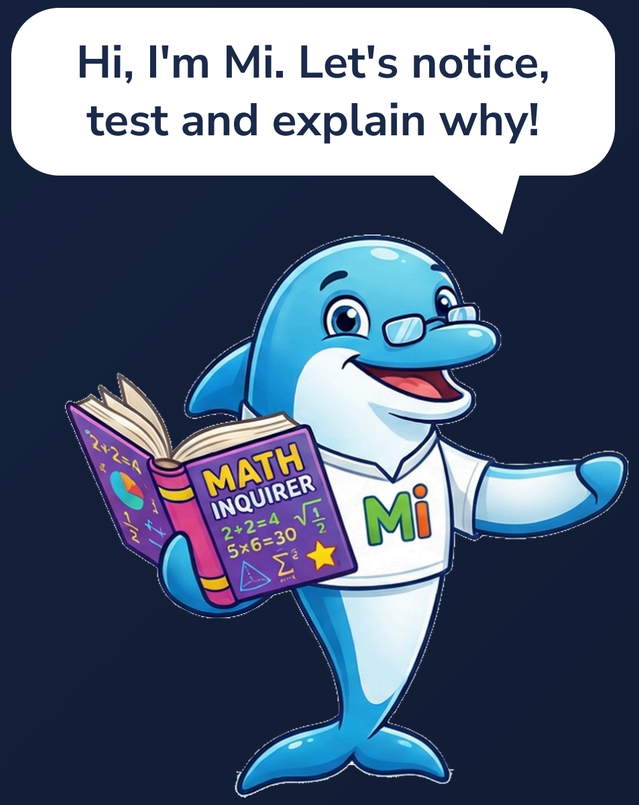



**TODAY WE WILL**

Meet seven tools.

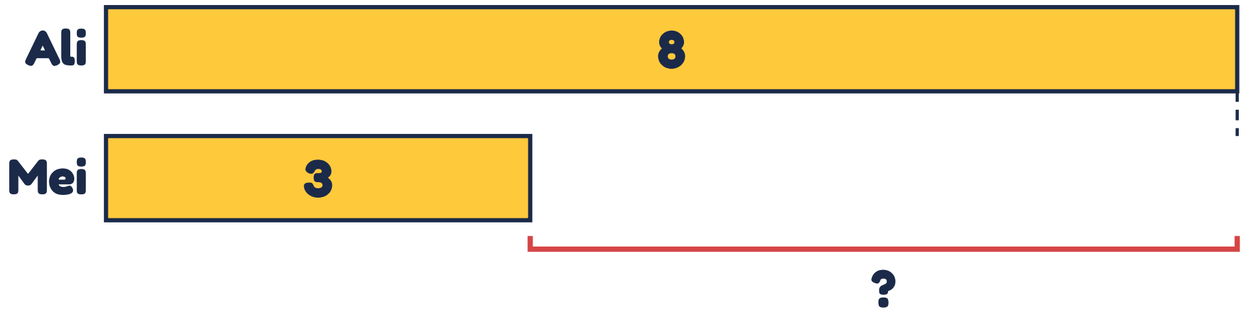
Pick the tool that fits.

Check an answer a second way.



# Warm-up: draw two strips

2–8 MIN



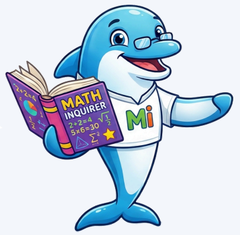

**Ali has 8 stickers. Mei has 3 stickers. How many more stickers does Ali have than Mei?**

# Find one first

2–8 MIN



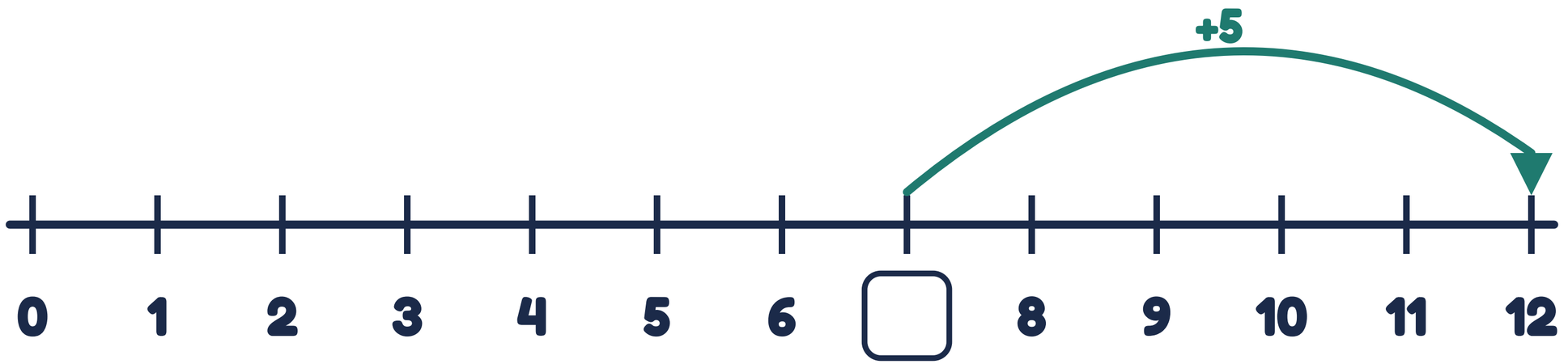
**3 pens cost $12. All the pens cost the same. How much does 1 pen cost?**




Take your time!

# Work backwards

2–8 MIN




I think of a number. I add 5. I get 12. What is my number?

# Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

## PUZZLE 1: anyone can try

Mei has 4 stickers.

Ali has 3 more stickers than Mei.

How many stickers does Ali have?

## PUZZLE 2: think

Mei and Ali have 10 stickers in all.

Ali has 2 more stickers than Mei.

How many stickers does Mei have?

I know because ___.

# Explore: your seven tools

12–23 MIN

| | | | |
|---|---|---|---|
| Strip | Table | List | Draw it |
| Work backwards | Find one first | Steps | |

Lay your 7 tool cards in a row. Which tool have you used before?

# Explore: problems 1 to 4

12–23 MIN

| Card | Problem |
|---|---|
| 1 | Ali has 9 pens. Mei has 4 fewer pens than Ali. How many pens do they have in all? |
| 2 | Rows of blocks have 2, 4, 6 blocks, and so on. Each row has 2 more than the row before. How many blocks are in the 6th row? |
| 3 | Mei picks 2 fruits from an apple, a pear and a plum. List every pair she can pick. |
| 4 | Ravi is 3rd in a line of 7 children. How many children are behind Ravi? |

**Put each problem card under a tool you would use. Say why.**

12–23 MIN

# A tool that fits: 1 to 4

| Card | A tool that fits |
|---|---|
| 1 | Strip |
| 2 | Table |
| 3 | List |
| 4 | Draw it |

**More than one tool can fit. A good reason matters more than the name.**

# Explore: problems 5 to 8

12–23 MIN

| Card | Problem |
|---|---|
| 5 | A number take away 6 is 9. What is the number? |
| 6 | 5 buns cost $15. All the buns cost the same. How much do 8 buns cost? |
| 7 | Ben has 20 sweets. He eats 8 sweets. Then he buys 5 more sweets. How many sweets does he have now? |
| 8 | For every 2 red beads there are 3 blue beads. How many blue beads go with 8 red beads? |

**Put each problem card under a tool you would use. Say why.**

12–23 MIN

# A tool that fits: 5 to 8

| Card | A tool that fits |
| --- | --- |
| 5 | Work backwards |
| 6 | Find one first |
| 7 | Steps |
| 8 | Table |

**Problems 2 and 8 both use a table. Many tools can fit one problem.**

# What did we notice?

12–23 MIN

- Does one word, like more, tell us the tool?
- Can two tools fit the same problem?

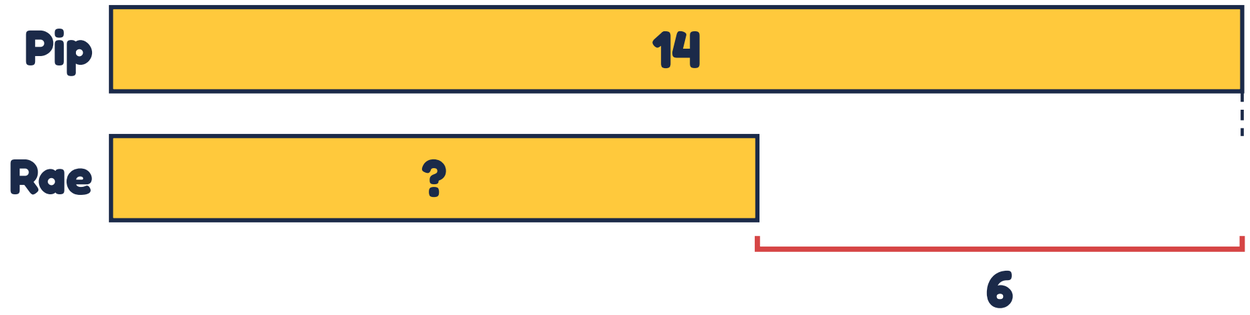
# Seven tools, seven clues

23–35 MIN

| | Tool | Use it when |
|---|---|---|
|  | Strip | Compare two amounts, or find a total. |
|  | Table | Numbers grow in a pattern. |
|  | List | Find every way. |
|  | Draw it | A line, a place or a shape. |
|  | Work backwards | You know the end, not the start. |
|  | Find one first | Equal groups, or the price of one. |
|  | Steps | One thing happens, then the next. |

# Strip: more and fewer

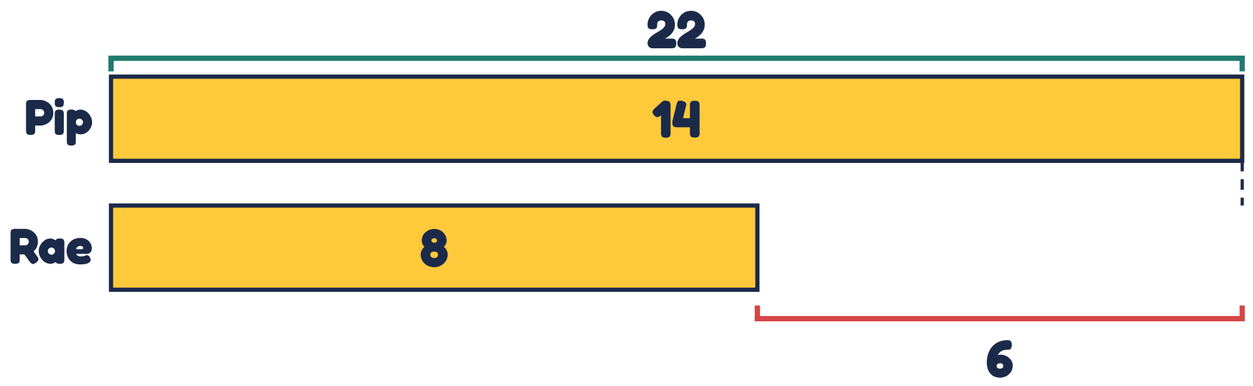
23–35 MIN




**Pip has 14 cards. Rae has 6 fewer cards than Pip. How many cards do they have in all?**

# Rae has 8 cards, 22 in all

23–35 MIN




**14 take away 6 is 8 cards for Rae. 14 + 8 = 22 cards in all.**

# Table: a growing pattern

23–35 MIN

| Path | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Tiles | 2 | 5 | 8 | | | | |

A path has 2 tiles, then 5 tiles, then 8 tiles. Each path has 3 more tiles than the path before. How many tiles are in the 7th path?

# 20 tiles in the 7th path

23–35 MIN

| Path | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Tiles | 2 | 5 | 8 | 11 | 14 | 17 | 20 |

**Add 3 each time: 2, 5, 8, 11, 14, 17, 20. The 7th path has 20 tiles.**

# Work backwards

23–35 MIN

| | | | | |
|---|---|---|---|---|
| ? | + 9 | | − 4 | 20 |

I think of a number. I add 9. Then I take away 4. I get 20. What is my number?

# My number is 15

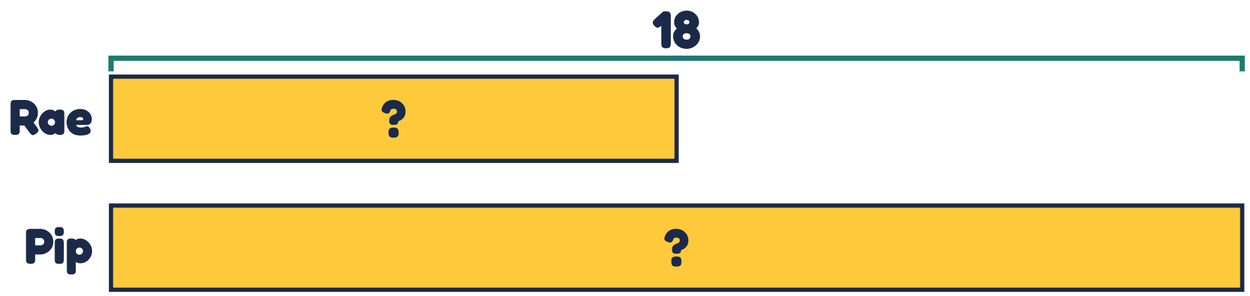
23–35 MIN

| 15 | + 9 | 24 | − 4 | 20 |
|---|---|---|---|---|

**Start at 20 and undo each step: 20 + 4 = 24, then 24 take away 9 is 15. Check: 15 + 9 = 24 and 24 take away 4 is 20.**

# Two roads, one answer

23–35 MIN




Pip has twice as many marbles as Rae. They have 18 marbles in all. How many marbles does Rae have?

# Road 2: a table of tries

23–35 MIN

| Rae | Pip | In all |
|---|---|---|
| 4 | 8 | 12 |
| 5 | 10 | 15 |
| 6 | 12 | 18 |

**Road 1, the strip: 3 equal parts, 18 shared into 3 is 6. Road 2, the table: Rae 6 and Pip 12 make 18. Both roads say Rae has 6 marbles.**

35–39 MIN

# Mime the tools

Stand up. Mime each tool with me.

Draw a long strip in the air. Tap a table.

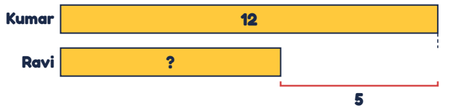
Sip some water.

# Workbook page 1: Start here

39–70 MIN

Questions 1 to 3. Take your time.

**S1** Use a strip. Kumar has 12 marbles. Ravi has 5 fewer marbles than Kumar.




**S2** Do one step at a time. A shop has 30 apples. It sells 12 apples in the morning and 9 apples in the afternoon.

**S3** Make a table. Each row has 3 more beads than the row before. Fill in the table.

| Row | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Beads | 3 | 6 | 9 | | | |

39–70 MIN

# Check page 1

| | |
|---|---|
| S1 | 7 marbles; 19 marbles in all |
| S2 | 18 apples; 9 apples |
| S3 | 15 beads; 18 beads |

# Workbook page 2: Start here

39–70 MIN

Questions 4 to 6. Take your time.

**S4** Work backwards. Ravi thinks of a number. He takes away 8 and gets 17.

**S5** Find one first. 4 bags hold 20 apples. All the bags hold the same number of apples.



**S6** Use a list. Ali has three coins: 5¢, 10¢ and 20¢. He picks 2 coins. Fill in the blanks.




PLAY MONEY

39–70 MIN

# Check page 2

| | |
|---|---|
| S4 | 25; 17 |
| S5 | 5 apples; 35 apples |
| S6 | 15¢, 25¢, 30¢; 3 pairs |

39–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | The tool is not named: you choose, and say why |
| **Go further (page 4)** | Two Roads, One Answer, make three amounts equal, a pattern |
| **Investigation (page 5)** | Pick a tool, write a problem |
| **Check** | Finish and check your own work |

**Everyone may choose. Not finished? Pages 3 to 5 are for home too.**

# Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

Pip has twice as many marbles as Rae.

They have 24 marbles in all.

24 shared into 2 is 12, so Rae has 12 marbles.

Do you agree? How could we check a second way?

I think … because …

# Back to the sneak-peek

70–76 MIN

**Puzzle 1: 7 stickers. Puzzle 2: Mei has 4 stickers. Mei 4 and Ali 6 make 10, and 6 is 2 more than 4.**

# Exit ticket

76–82 MIN

On your own. No looking back.

**E1** **Mei thinks of a number. She adds 4 and gets 15.**

Circle the tool that fits best.

**E2** **Use a strip. Ali has 11 stickers. Mei has 4 fewer stickers than Ali.**

(a) Mei has ☐ stickers.

(b) They have ☐ stickers in all.

**E3** **5 buns cost $20. A child in another class says 1 bun costs $5.**

Circle True or False.

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | Work backwards |
| E2 | 7 stickers; 18 stickers |
| E3 | False |

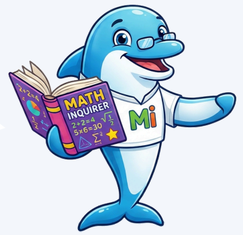
# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ picking a tool ☐ using a strip ☐ working backwards ☐ something else |
| **What helped me** | ☐ the tool cards ☐ drawing it ☐ checking a second way<br>☐ I was not stuck |
| **Next time I will try** | ☐ picking a tool ☐ checking a second way ☐ saying why<br>☐ asking a question |
| | ☐ I am still stuck (my grown-up will photograph this page for my tutor)<br>☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

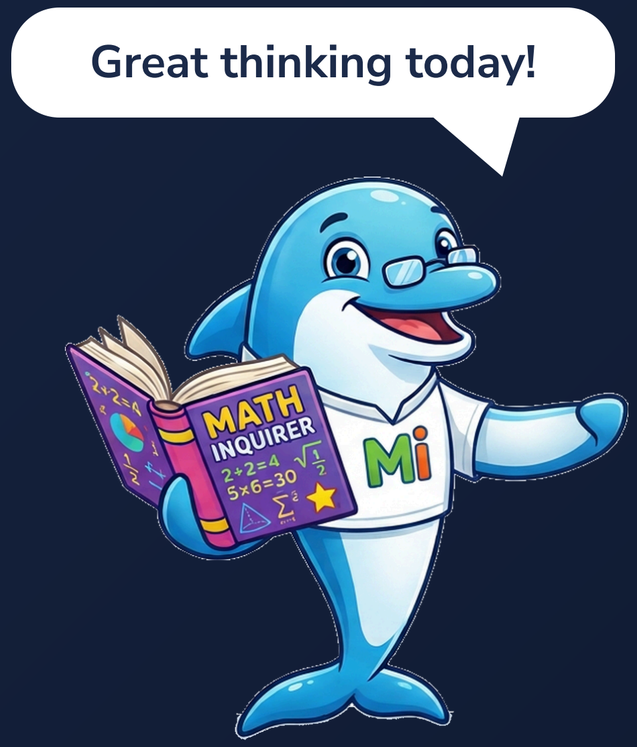
# Well done, inquirers!

Show a grown-up your tool cards. Say which tool you like best.

NEXT LESSON

Inquirers' Showcase and Mi's Challenge Quest

Have ready: your Quest Log, closed in its envelope, a quest partner, a favourite page from your book and plain paper.



# Go further: how many cuts?

OPTIONAL

**A rope is cut into 5 pieces. How many cuts are there? Draw it.**

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Can two different tools solve the same problem? Try one. |
| **Make a puzzle** | Pick a tool card and write a problem for your grown-up. |
| **Teach it back** | Tell a toy why we check an answer a second way. |

**Optional. Everyone may join.**